

Signals Project

Signals and Systems, 4CA20, 2025-2026



Project description

Design and implement an application of Signals and Systems in which the Discrete Fourier Transform is a central analytical tool. In addition, the project must include a structured analysis of parameter sensitivity, limitations, and failure modes of the DFT, supported by quantitative experiments.

Rules

- Your project must clearly demonstrate how the Discrete Fourier Transform (DFT) is essential to solving the problem you choose. The DFT should not just be used as a minor step but should be the core analytical tool in your project.
- You can use any programming language (MATLAB, Python, C, etc.); MATLAB or Python are recommended.
- You can solve the project in groups of at most 3 people.
- The project must include a structured robustness analysis of the signal using the Discrete Fourier Transform. From the list of stress tests below, each group must select **at least two** tests that are meaningful for their chosen application and apply them.
 - **Parameter sensitivity.** Systematically vary at least one key DFT-related parameter (e.g. observation window length, DFT size, or frequency resolution) and analyze the effect of this variation.
 - **Sampling or resolution.** Investigate the effect of sampling rate or effective resolution, for example by downsampling, and identify any resulting loss of information or aliasing effects.
 - **Noise robustness.** Add controlled noise to the signal and analyze the impact on the DFT based conclusions, explicitly stating which conclusions remain valid and which do not.
 - **Failure or limitation case.** Identify one scenario in which the DFT provides a misleading, incomplete, or ambiguous interpretation of the signal, and explain the reason.
- The deliverables are (submit them on Canvas under assignments in one .zip file with group number, e.g. group22.zip):
 - **Video** (max. duration 4 minutes) explaining your application. The video must include signal theoretic reasoning linking the observed phenomena to concepts from the Signals lectures, including robustness, parameter sensitivity, and at least one identified limitation or failure case of the DFT. The video must feature your group presenting your own work.
 - **Code and instruction files** to test your concept. If your concept involves real data (e.g., voice, image), provide the corresponding data. Include a “run_me” file that generates all figures and results used for analysis, including those related to parameter variation and stress testing.
 - **Reflection on AI assisted reasoning (max. 1 page)** critically assessing the use of automated tools in the project. The reflection must include: 1) which tools were used, 2) how and why they were used, 3) at least one limitation, incorrect assumption, or misleading output and an explanation of why it is incorrect or insufficient, and 4) which tasks the tools were not used for and why.

Evaluation

- Deadline: All the deliverables should be submitted via Canvas before **June 1st, 2026, 23h59**.
- The project will be evaluated based on four items:
 1. Idea and appropriateness of the application;
 2. Interpretation and reasoning;
 3. Quality and clarity of the video;
 4. Reproducibility, code quality, and critical reflection.

Each category is graded from 0-5, and your total score is divided by 2 to yield a final score out of 10.

- The project will be evaluated by 2 assessors and average of the two grades will be taken per item.

Rubric

The item-by-item rubric of evaluation is presented below.

Idea and appropriateness of the application. Points of interest: Is the chosen application well matched to the DFT? Is the DFT crucial to solving the problem? Are the goals of the application clearly defined and meaningful?

- 0-1 The application closely follows examples provided in the course or uses the DFT in a superficial way that is not essential to the analysis.
- 2-3 The application uses the DFT meaningfully, but its relevance is only partially justified.
- 4-5 The idea is original and DFT is essential in solving the problem (i.e., no other tool would be as effective).

Interpretation and reasoning. Points of interest: Are conclusions supported by quantitative evidence? Are parameter sensitivity, robustness, and limitations correctly identified and interpreted? Are the selected stress tests appropriate and well justified?

- 0-1 Results are presented with little or no interpretation, or interpretations are incorrect or inconsistent.
- 2-3 Some interpretation is provided and stress tests are performed, but explanations are partial, generic, or lack clear connection to Signals.
- 4-5 Results are carefully interpreted, with clear discussion and reasoning of two or more stress tests.

Quality and clarity of the video. Points of interest: Is the video clear and well structured? Does it focus on explaining results and interpretations rather than code walkthroughs? Are figures and visualizations used effectively to support the explanation?

- 0-1 The video lacks structure, explanations are unclear, or the audio or visual quality prevents understanding.
- 2-3 The video explains the results, but clarity, structure or interpretation could be improved.
- 4-5 The video is well organized, clearly explains the results and their interpretation, and uses visual elements effectively to support understanding.

Reproducibility, code quality, and critical reflection. Points of interest: Is the analysis reproducible from the provided code? Is the code consistent with Signals concepts from the course? Is there evidence of critical assessment of automated assistance?

- 0-1 Code does not run, or does not reproduce the results shown. Little or no critical reflection is provided.
- 2-3 Code runs and reproduces most results, but organization, documentation, or conceptual clarity is limited. The critical reflection is present but superficial.
- 4-5 Code is functional, well organized, and reproducible, with clear links to Signals. The reflection correctly identifies and explains a limitation or misleading automated output.

Examples

Videos of selected top 10 projects of previous years are available here:

- Year 2024: <https://youtu.be/hnAp-CB2F0M>
- Year 2025: <https://youtu.be/HvR9xX06xLs>

Note that these videos do not necessarily include stress tests. For guidance on the selection of the application, or the implementation itself, please contact the teaching assistants of your guided self study group. If any point of this document is unclear, please submit your point of clarification on the discussion page in Canvas.